



LAXMI CHARITABLE TRUST'S
SHETH L.U.J. COLLEGE OF ARTS & SIR M.V. COLLEGE OF
SCIENCE & COMMERCE
DR. S. RADHAKRISHNAN MARG, ANDHERI(EAST), MUMBAI - 400069

PRACTICAL NO 3

AIM: Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

NAME: KRISHNA NANDKUMAR CHAUDHARY
ROLL NO.: T074
SUBJECT: CYBER SECURITY



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CODE CLI:

```
T074 Cyber Sec Pract 3.py - C:/Users/PRIYANKA/Downloads/T074 Cyber Sec Pract 3.py (3.13.1)
File Edit Format Run Options Window Help
#Krishna T074 Cyber Security Pract 3

import hmac
import hashlib

# Function to generate MAC
def generate_mac(message, secret_key):
    mac = hmac.new(
        secret_key.encode(),
        message.encode(),
        hashlib.sha256
    )
    return mac.hexdigest()

# Function to verify MAC
def verify_mac(message, secret_key, received_mac):
    generated_mac = generate_mac(message, secret_key)
    return hmac.compare_digest(generated_mac, received_mac)

# Main Program
message = input("Enter the message: ")
secret_key = input("Enter the secret key: ")

# Generate MAC
mac = generate_mac(message, secret_key)
print("\nGenerated MAC:", mac)

# Verify MAC
print("\n--- Verification ---")
received_message = input("Enter the received message: ")
received_mac = input("Enter the received MAC: ")

if verify_mac(received_message, secret_key, received_mac):
    print("MAC Verification Successful!")
    print("Message is authentic and has not been modified.")
else:
    print("MAC Verification Failed!")
    print("Message has been altered or the key is incorrect.")

Ln: 21 Col: 0
```

OUTPUT CLI:

```
C:\Windows\system32\cmd.e. x + v
Microsoft Windows [Version 10.0.26280.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\PRIYANKA>python "C:\Users\PRIYANKA\Downloads\T074 Cyber Sec Pract 3.py"
Enter the message: there is no pract today
Enter the secret key: nopract11

Generated MAC: b29553f59eed66065862c3dffc164cf7905ab43cc861e443befd602b4e3c4a54

--- Verification ---
Enter the received message: there is no pract today
Enter the received MAC: b29553f59eed66065862c3dffc164cf7905ab43cc861e443befd602b4e3c4a54
MAC Verification Successful!
Message is authentic and has not been modified.

C:\Users\PRIYANKA>
```

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CODE GUI:

```
T074 Cyber Sec Pract 3.py - C:/Users/PRIVANKA/Downloads/T074 Cyber Sec Pract 3.py (3.13.1)
File Edit Format Run Options Window Help
# Krishna T074 Cyber Security Pract 3 (GUI)

import tkinter as tk
from tkinter import messagebox
import hmac
import hashlib

# Function to generate MAC
def generate_mac(message, secret_key):
    mac = hmac.new(
        secret_key.encode(),
        message.encode(),
        hashlib.sha256
    )
    return mac.hexdigest()

# Generate MAC Button
def generate():
    message = message_entry.get()
    secret_key = key_entry.get()

    if message == "" or secret_key == "":
        messagebox.showerror("Error", "Please enter Message and Secret Key.")
        return

    mac = generate_mac(message, secret_key)
    mac_entry.delete(0, tk.END)
    mac_entry.insert(0, mac)

# Verify MAC Button
def verify():
    received_message = received_message_entry.get()
    secret_key = key_entry.get()
    received_mac = received_mac_entry.get()

    if received_message == "" or received_mac == "":
        messagebox.showerror("Error", "Please fill all verification fields.")
        return

    generated_mac = generate_mac(received_message, secret_key)
```

```
T074 Cyber Sec Pract 3.py - C:/Users/PRIVANKA/Downloads/T074 Cyber Sec Pract 3.py (3.13.1)
File Edit Format Run Options Window Help

    generated_mac = generate_mac(received_message, secret_key)

    if hmac.compare_digest(generated_mac, received_mac):
        messagebox.showinfo("Success", "MAC Verification Successful!\n\nMessage is authentic and has not been modified.")
    else:
        messagebox.showerror("Failed", "MAC Verification Failed!\n\nMessage has been altered or the key is incorrect.")

# GUI Window
root = tk.Tk()
root.title("Message Authentication Code (MAC)")
root.geometry("700x450")
root.resizable(False, False)

# Labels and Entries
tk.Label(root, text="Message:", font=("Arial", 11)).pack(pady=5)
message_entry = tk.Entry(root, width=70)
message_entry.pack()

tk.Label(root, text="Secret Key:", font=("Arial", 11)).pack(pady=5)
key_entry = tk.Entry(root, width=70, show="*")
key_entry.pack()

tk.Button(root, text="Generate MAC", command=generate, bg="green", fg="white").pack(pady=10)

tk.Label(root, text="Generated MAC:", font=("Arial", 11)).pack()
mac_entry = tk.Entry(root, width=70)
mac_entry.pack()

tk.Label(root, text="Received Message:", font=("Arial", 11)).pack(pady=10)
received_message_entry = tk.Entry(root, width=70)
received_message_entry.pack()

tk.Label(root, text="Received MAC:", font=("Arial", 11)).pack(pady=5)
received_mac_entry = tk.Entry(root, width=70)
received_mac_entry.pack()

tk.Button(root, text="Verify MAC", command=verify, bg="blue", fg="white").pack(pady=15)

root.mainloop()
```

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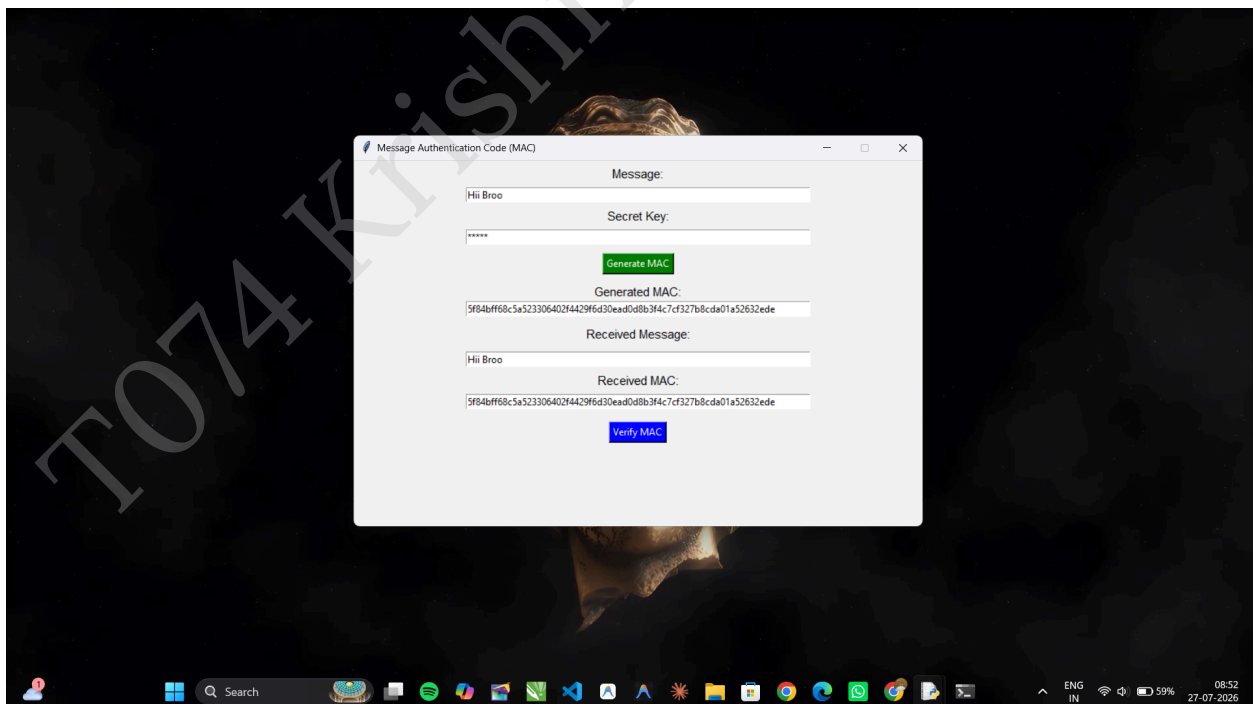
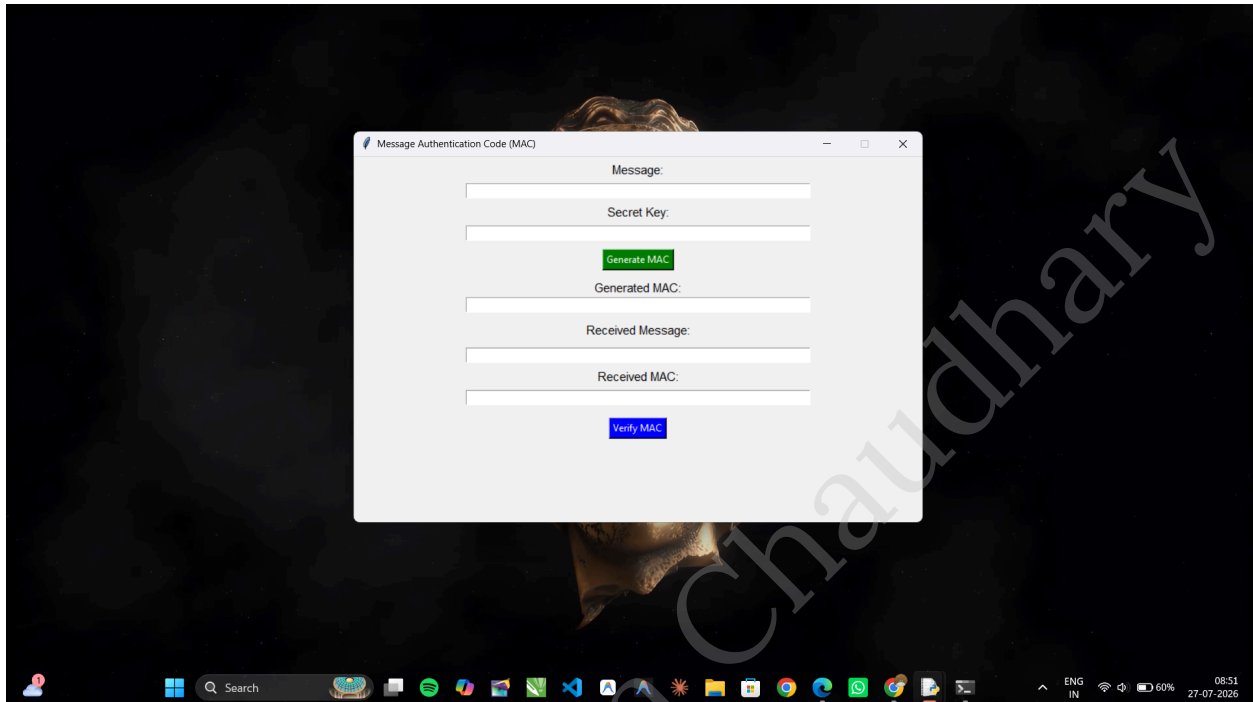
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OUTPUT GUI:



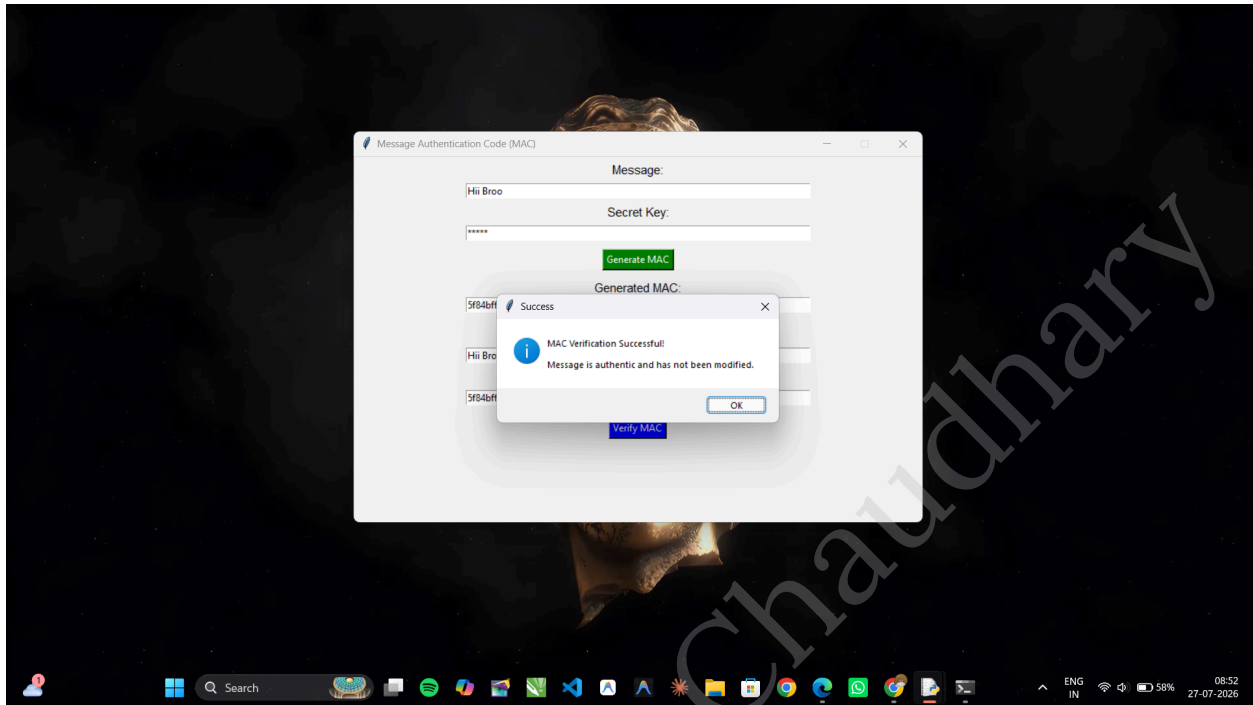
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